

CHALLENGER EXERCISES

Exercise C - I

1. If $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$, then the value(s) of $\cos\left(\theta - \frac{\pi}{4}\right)$ is (are) :

- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\pm \frac{1}{2\sqrt{2}}$ (D) None of these

यदि $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$ तब $\cos\left(\theta - \frac{\pi}{4}\right)$ का मान ज्ञात होगा :

- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\pm \frac{1}{2\sqrt{2}}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$

$$\Rightarrow \tan(\pi \cos \theta) = \tan\left(\frac{\pi}{2} - \pi \sin \theta\right)$$

$$\Rightarrow \pi \cos \theta = n\pi + \frac{\pi}{2} - \pi \sin \theta$$

$$\Rightarrow \cos \theta + \sin \theta = \frac{(2n+1)}{2} \Rightarrow \cos\left(\theta - \frac{\pi}{4}\right) = \left(\frac{2n+1}{2\sqrt{2}}\right)$$

$$\text{For } n = 0 \quad \cos\left(\theta - \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$$

$$\text{For } n = 1 \quad \cos\left(\theta - \frac{\pi}{4}\right) = \frac{3}{2\sqrt{2}}$$

$$\text{For } n = -1 \quad \cos\left(\theta - \frac{\pi}{4}\right) = \frac{-1}{2\sqrt{2}}$$

2. The equation, $\sin^4 x - 2 \cos^2 x + a^2 = 0$ is solvable if:

- (A) $-\sqrt{3} \leq a \leq \sqrt{3}$ (B) $-\sqrt{2} \leq a \leq \sqrt{2}$ (C) $-1 \leq a \leq 1$ (D) None of these

समीकरण $\sin^4 x - 2 \cos^2 x + a^2 = 0$ का हल संभव है यदि :

- (A) $-\sqrt{3} \leq a \leq \sqrt{3}$ (B) $-\sqrt{2} \leq a \leq \sqrt{2}$ (C) $-1 \leq a \leq 1$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\sin^4 x - 2 \cos^2 x + a^2 = 0$

$$\Rightarrow \sin^4 x - 2(1 - \sin^2 x) + a^2 = 0$$

$$\Rightarrow \sin^4 x + 2 \sin^2 x + (a^2 - 2) = 0$$

$$\Rightarrow (\sin^2 x + 1)^2 + a^2 - 3 = 0$$

$$\Rightarrow a^2 = 3 - (\sin^2 x + 1)^2$$

$$\because \sin^2 x \in [0, 1]$$

$$\Rightarrow (\sin^2 x + 1) \in [1, 2]$$

$$\Rightarrow (\sin^2 x + 1)^2 \in [1, 4]$$

$$\Rightarrow 3 - (\sin^2 x + 1)^2 \in [-1, 2]$$

$$\Rightarrow 0 \leq a^2 \leq 2 \quad \Rightarrow -\sqrt{2} \leq a \leq \sqrt{2}$$

3. The general solution of the equation, $\frac{1 - \sin x + \dots + (-1)^n \sin^n x + \dots}{1 + \sin x + \dots + \sin^n x + \dots} = \frac{1 - \cos 2x}{1 + \cos 2x}$ is :

निम्न समीकरण $\frac{1 - \sin x + \dots + (-1)^n \sin^n x + \dots}{1 + \sin x + \dots + \sin^n x + \dots} = \frac{1 - \cos 2x}{1 + \cos 2x}$ का व्यापक हल है :

(A) $(-1)^n \left(\frac{\pi}{3}\right) + n\pi$ (B) $(-1)^n \left(\frac{\pi}{6}\right) + n\pi$ (C) $(-1)^{n+1} \left(\frac{\pi}{6}\right) + n\pi$ (D) $(-1)^{n-1} \left(\frac{\pi}{3}\right) + n\pi$, ($n \in \mathbb{I}$)

Ans. B

Sol. $\frac{1 - \sin x + \dots + (-1)^n \sin^n x + \dots}{1 + \sin x + \dots + \sin^n x + \dots} = \frac{1 - \cos 2x}{1 + \cos 2x}$

$$\Rightarrow \frac{1}{\frac{1 + \sin x}{1 - \sin x}} = \tan^2 x$$

$$\Rightarrow \frac{1 - \sin x}{1 + \sin x} = \tan^2 x$$

$$\Rightarrow (1 - \sin^2 x)(1 - \sin x) = (1 + \sin x) \sin^2 x$$

$$\Rightarrow 2 \sin^2 x + \sin x - 1 = 0$$

$$\Rightarrow (2 \sin x - 1)(\sin x + 1) = 0$$

$$\Rightarrow \underbrace{\sin x = -1}_{\text{Not possible as } \cos x \neq 0} \quad \text{or} \quad \sin x = \frac{1}{2}$$

$$\Rightarrow \sin x = \frac{1}{2}$$

$$\Rightarrow x = n\pi + (-1)^n \frac{\pi}{6}; n \in \mathbb{I}$$

4. If the equation, $x^2 + 4 + 3 \sin(ax + b) - 2x = 0$ has atleast one real solution, where $a, b \in [0, 2\pi]$, then one possible value of $(a + b)$ can be equal to :

(A) $\frac{7\pi}{2}$ (B) $\frac{5\pi}{2}$ (C) $\frac{9\pi}{2}$ (D) None of these

यदि समीकरण $x^2 + 4 + 3 \sin(ax + b) - 2x = 0$ के कम से कम एक वास्तविक हल है जहां $a, b \in [0, 2\pi]$ तब $(a + b)$ का एक सम्भव हल होगा :

- (A) $\frac{7\pi}{2}$ (B) $\frac{5\pi}{2}$ (C) $\frac{9\pi}{2}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $x^2 + 4 + 3 \sin(ax + b) - 2x = 0$

$$\Rightarrow \frac{1}{3}[-x^2 + 2x - 4] = \sin(ax + b)$$

$$\therefore \sin(ax + b) \in [-1, 1]$$

$$\& -\frac{x^2}{3} + \frac{2x}{3} - \frac{4}{3} = 0 \in \left(-\infty, -\frac{D}{4a}\right] \equiv (-\infty, -1]$$

Only possibility are $\underbrace{\sin(ax + b) = -1}_{\text{only for } x = 1}$

$$\Rightarrow a + b = (4n - 1)\frac{\pi}{2}$$

$$\Rightarrow a + b = \frac{7\pi}{2} \quad \{\because a, b \in [0, 2\pi]\}$$

5. The number of points of intersection of the two curves, $y = 2 \sin x$ and $y = 5x^2 + 2x + 3$, is :

- (A) Infinite (B) 0 (C) 1 (D) 2

दो वक्रों $y = 2 \sin x$ एवं $y = 5x^2 + 2x + 3$ के प्रतिच्छेदन से बने बिन्दुओं की संख्या है :

- (A) अन्त (B) 0 (C) 1 (D) 2

Ans. B

Sol. $\underbrace{y = 2 \sin x}_{\because 2 \sin x \in [-2, 2]} \quad \& \quad y = 5x^2 + 2x + 3$

$$D = 4 - 60 = -56 < 0$$

$$a = 5 > 0$$

$$-\frac{b}{2a} = \frac{-2}{10} = \frac{-1}{5}$$

$$-\frac{D}{4a} = \frac{56}{4 \times 5} = \frac{14}{5}$$

$$5x^2 + 2x + 3 \in \left[\frac{14}{5}, \infty\right)$$

$\therefore$ No. of points of intersection of the two curve is 0.

6. The number of all possible triplets (a_1, a_2, a_3) such that $a_1 + a_2 \cos(2x) + a_3 \sin^2(x) = 0$ for all x is :
 (A) zero (B) one (C) three (D) infinite

सभी x के लिए समीकरण $a_1 + a_2 \cos(2x) + a_3 \sin^2(x) = 0$ त्रिक (a_1, a_2, a_3) की संख्या होगी :

- (A) शून्य (B) एक (C) तीन (D) अनन्त

Ans. D

Sol. $a_1 + a_2 \cos(2x) + a_3 \sin^2(x) = 0 \quad \forall x \in \mathbb{R}$
 $\Rightarrow a_1 + a_2 + \sin^2 x (a_3 - 2a_2) = 0 \quad \forall x \in \mathbb{R}$
 $\Rightarrow a_1 + a_2 = 0 \quad \& \quad a_3 - 2a_2 = 0$
 $\Rightarrow (a_1, a_2, a_3) = (-a_2, a_2, 2a_2) = a_2(-1, 1, 2)$
 $\therefore$ There are infinite triplets.

7. The number of values of x in the interval $(0, 5\pi)$, satisfying the equation $3 \sin^2 x - 7 \sin x + 2 = 0$ is :

अन्तराल $(0, 5\pi)$ में समीकरण $3 \sin^2 x - 7 \sin x + 2 = 0$ के हलों की संख्या है :

- (A) 0 (B) 5 (C) 6 (D) 10

Ans. C

Sol. Given $x \in (0, 5\pi)$
 $3 \sin^2 x - 7 \sin x + 2 = 0$
 $\Rightarrow (3 \sin x - 1)(\sin x - 2) = 0$
 $\Rightarrow \underbrace{\sin x = \frac{2}{3}}_{\text{Not possible}} \quad \text{or} \quad \sin x = \frac{1}{3}$
 $\Rightarrow \sin x = \frac{1}{3}$; In $(0, 5\pi)$ No. of values of x is 6.

8. If $\sin^2 \theta - \sin \theta - 1 = 0$ is to be satisfied for exactly 4 distinct value of $\theta \in [0, n\pi]$, $n \in \mathbb{N}$, then least value of n is:

यदि समीकरण $\sin^2 \theta - \sin \theta - 1 = 0$ चार अलग-अलग मानों के लिए सन्तुष्ट होती है तथा $\theta \in [0, n\pi]$, $n \in \mathbb{N}$, तब n का न्यूनतम मान होगा :

- (A) 2 (B) 6 (C) 4 (D) 1

Ans. C

Sol. $\sin^2 \theta - \sin \theta - 1 = 0$

$$\Rightarrow \sin \theta = \frac{1 \pm \sqrt{5}}{2}$$

$$\Rightarrow \underbrace{\sin \theta = \frac{1 + \sqrt{5}}{2}}_{\text{Not possible}} \quad \text{or} \quad \sin \theta = \frac{1 - \sqrt{5}}{2}$$

In $[0, 4\pi]$ No. of distinct values of θ is 4.

$\therefore n = 4$

9. If $\tan(\cot x) = \cot(\tan x)$, then $\sin 2x =$

यदि $\tan(\cot x) = \cot(\tan x)$ तब $\sin 2x =$

(A) $\frac{4}{(2n+1)\pi}; n \in I$

(B) $\frac{4}{(2n+1)\pi}; n \in I - \{-1, 0\}$

(C) $\frac{4}{(2n+1)\pi}; n \in I - \{-1\}$

(D) $\frac{4}{(2n+1)\pi}; n \in I - \{0\}$

Ans. B

Sol. $\tan(\cot x) = \cot(\tan x)$

$$\Rightarrow \tan(\cot x) = \tan\left(\frac{\pi}{2} - \tan x\right)$$

$$\Rightarrow \cot x = n\pi + \frac{\pi}{2} - \tan x$$

$$\Rightarrow \tan x + \cot x = n\pi + \frac{\pi}{2}$$

$$\Rightarrow \frac{2}{\sin 2x} = (2n+1)\frac{\pi}{2}$$

$$\Rightarrow \sin 2x = \frac{4}{(2n+1)\pi}; n \in Z - \{-1, 0\}$$

10. The general solution of the equation $\tan^2 \alpha + 2\sqrt{3} \tan \alpha = 1$ is given by :

समीकरण $\tan^2 \alpha + 2\sqrt{3} \tan \alpha = 1$ का व्यापक हल है :

(A) $\alpha = \frac{n\pi}{2}$ (B) $\alpha = (2n+1)\frac{\pi}{2}$ (C) $\alpha = (6n+1)\frac{\pi}{12}$ (D) $\alpha = n\frac{\pi}{12}$

Ans. C

Sol. $\tan^2 \alpha + 2\sqrt{3} \tan \alpha - 1 = 0$

$$\tan \alpha = \frac{-2\sqrt{3} \pm 4}{2}$$

$$\Rightarrow \tan \alpha = -(2 + \sqrt{3}) \quad \text{or}$$

$$\tan \alpha = 2 - \sqrt{3}$$

$$\Rightarrow \tan \alpha = \tan\left(\frac{\pi}{2} + \frac{\pi}{12}\right)$$

$$\text{or } \tan \alpha = \tan\left(\frac{\pi}{12}\right)$$

$$\Rightarrow \alpha = n\pi + \frac{7\pi}{12}$$

$$\text{or } \alpha = n\pi + \frac{\pi}{12}$$

$$\Rightarrow \alpha = (6n+1) \frac{\pi}{12}$$

11. Number of real values of x in $[-2\pi, 2\pi]$ for which value of the expression $y = \sqrt{2\sin x} - \sin x$ is maximum, is :

(A) 0 (B) 2 (C) 4 (D) more than 4

अन्तराल $[-2\pi, 2\pi]$ में x के उन वास्तविक मानों की संख्या, जिनके लिए व्यंजक $y = \sqrt{2\sin x} - \sin x$ का मान अधिकतम है, होगी :

(A) 0 (B) 2 (C) 4 (D) 4 से अधिक

Ans. C

Sol. $x \in [-2\pi, 2\pi]$

$$y = \sqrt{2\sin x} - \sin x$$

$$\text{Let } \sqrt{\sin x} = t$$

$$\Rightarrow \sin x = t^2$$

$$\Rightarrow y = \sqrt{2}t - t^2$$

y is maximum at Abscissa value of vertex.

$$\therefore \text{Abscissa value of vertex} = \frac{-b}{2a} = \frac{-\sqrt{2}}{-2 \times 1} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow t = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \sqrt{\sin x} = \frac{1}{\sqrt{2}} \Rightarrow \sin x = \frac{1}{2}$$

In $[-2\pi, 2\pi]$ no. of real values of x is 4.

$$\therefore n = 4$$

12. The set of all x in $(-\pi, \pi)$ satisfying $|4\sin x - 1| < \sqrt{5}$ is given by :

(A) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (B) $\left(\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (C) $\left(\frac{\pi}{10}, -\frac{3\pi}{10}\right)$ (D) none of these

अन्तराल $(-\pi, \pi)$ में असमिका $|4\sin x - 1| < \sqrt{5}$ को संतुष्ट करने वाले x के सभी मानों का समुच्चय होगा :

(A) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (B) $\left(\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (C) $\left(\frac{\pi}{10}, -\frac{3\pi}{10}\right)$ (D) इनमें से कोई नहीं

Ans. A

Sol. $|4\sin x - 1| < \sqrt{5} \Rightarrow -\sqrt{5} < 4\sin x - 1 < \sqrt{5}$

$$\Rightarrow -\left(\frac{\sqrt{5}-1}{4}\right) < \sin x < \left(\frac{\sqrt{5}+1}{4}\right)$$

$$\Rightarrow -\sin \frac{\pi}{10} < \sin x < \sin \frac{3\pi}{10}$$

$$\Rightarrow \sin\left(-\frac{\pi}{10}\right) < \sin x < \sin \frac{3\pi}{10}$$

$$\Rightarrow x \in \left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$$

13. The values of x between 0 and 2π , which satisfy the equation $\sin x \sqrt{8\cos^2 x} = 1$ are in A.P. with common difference:

0 से 2π के मध्य x के मान, जो समीकरण $\sin x \sqrt{8\cos^2 x} = 1$ को संतुष्ट करते हैं, समान्तर श्रेणी में हैं तो सार्वअन्तर होगा :

- (A) $\pi/4$ (B) $\pi/8$ (C) $3\pi/8$ (D) $5\pi/8$

Ans. A

Sol. $x \in (0, 2\pi)$

$$\sin x \sqrt{8\cos^2 x} = 1$$

$$2\sin x |\cos x| = \frac{1}{\sqrt{2}}$$

Case I :- If $\cos x \geq 0$

$$\text{then } \sin 2x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow 2x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

$$\therefore \cos x > 0$$

$$\Rightarrow x = \frac{\pi}{8}, \frac{3\pi}{8}$$

Case II :- If $\cos x < 0$ then $\sin 2x = -\frac{1}{\sqrt{2}}$

$$\Rightarrow 2x = \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4}$$

$$\therefore \cos x < 0$$

$$\Rightarrow x = \frac{5\pi}{8}, \frac{7\pi}{8}$$

$$\therefore x = \underbrace{\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}}_{\substack{\text{A.P. with common} \\ \text{difference} = \frac{\pi}{4}}}$$

14. α, β, γ and δ are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity k . The value of $4 \sin \frac{\alpha}{2} + 3 \sin \frac{\beta}{2} + 2 \sin \frac{\gamma}{2} + \sin \frac{\delta}{2}$ is equal to :

यदि α, β, γ तथा δ बढ़ते हुये क्रम में न्यूनतम धनात्मक कोण हो जहाँ सभी के sines का मान समान है और k के बराबर है तो

$4 \sin \frac{\alpha}{2} + 3 \sin \frac{\beta}{2} + 2 \sin \frac{\gamma}{2} + \sin \frac{\delta}{2}$ का मान होगा :

- (A) $2\sqrt{1-k}$ (B) $2\sqrt{1+k}$ (C) $2\sqrt{k}$ (D) $2k$

Ans. B

Sol. $\because \alpha < \beta < \gamma < \delta$ and $\sin \alpha = \sin \beta = \sin \gamma = \sin \delta = k$
 $\Rightarrow \alpha, \beta = \pi - \alpha, \gamma = 2\pi + \alpha, \delta = 3\pi - \alpha$

$$\Rightarrow y = 4 \sin \frac{\alpha}{2} + 3 \sin \frac{\beta}{2} + 2 \sin \frac{\gamma}{2} + \sin \frac{\delta}{2}$$

$$\Rightarrow y = 4 \sin \frac{\alpha}{2} + 3 \sin \left(\frac{\pi - \alpha}{2} \right) + 2 \sin \left(\pi + \frac{\alpha}{2} \right) + \sin \left(\frac{3\pi - \alpha}{2} \right)$$

$$\Rightarrow y = 4 \sin \frac{\alpha}{2} + 3 \cos \frac{\alpha}{2} - 2 \sin \frac{\alpha}{2} - \cos \frac{\alpha}{2}$$

$$\Rightarrow y = 2 \left(\sin \frac{\alpha}{2} + \cos \frac{\alpha}{2} \right)$$

$$\Rightarrow y^2 = 4(1 + \sin \alpha)$$

$$\Rightarrow y^2 = 4(1 + k)$$

$$\Rightarrow y = 2\sqrt{1+k}$$

15. The number of points inside or on the circle $x^2 + y^2 = 4$ satisfying $\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$ is :

- (A) one (B) two (C) four (D) infinite

वृत्त $x^2 + y^2 = 4$ पर या अन्दर स्थित बिन्दुओं की संख्या, जो $\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$ को संतुष्ट करते हैं :

- (A) एक (B) दो (C) चार (D) अनन्त

Ans. C

Sol. $x^2 + y^2 = 4$ & $\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$

$$\therefore \tan^4 x + \cot^4 x + 1 \in [3, \infty)$$

$$\& 3 \sin^2 y \in [0, 3]$$

$$\begin{aligned} \Rightarrow 3 \sin^2 y &= 3 & \& \tan^4 x + \cot^4 x + 1 = 3 \\ \Rightarrow \sin^2 y &= 1 & \& \tan^4 x + \cot^4 x = 2 \\ \Rightarrow y &= (2n+1)\frac{\pi}{2} & \& \tan^4 x = 1 \\ \Rightarrow y &= (2n+1)\frac{\pi}{2} & \& x = n\pi \pm \frac{\pi}{4} \\ \Rightarrow y &= -\frac{\pi}{2}, \frac{\pi}{2} & \& x = \frac{\pi}{4}, -\frac{\pi}{4} & \left\{ \because x^2 + y^2 = 4 \right\} \\ \Rightarrow \text{No. of pairs are 4.} \end{aligned}$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

16. $|\cos x| = \cos x - 2 \sin x$, if :

$|\cos x| = \cos x - 2 \sin x$, यदि :

- (A) $x = n\pi$ (B) $x = 2n\pi$ (C) $x = n\pi + \frac{\pi}{4}$ (D) $x = (2n+1)\pi + \frac{\pi}{4}$

Ans. BD

Sol. $|\cos x| = \cos x - 2 \sin x$

Case I :- If $\cos x \geq 0$ then

$$\sin x = 0$$

$$\Rightarrow x = n\pi; n \in \mathbb{I}$$

$$\Rightarrow x = 0, \pi, 2\pi, 3\pi, \dots$$

$$\because \cos x \geq 0$$

$$\Rightarrow x = 0, 2\pi, 4\pi, \dots$$

$$\Rightarrow x = 2n\pi; n \in \mathbb{I}$$

Case II :- If $\cos x < 0$

then $\tan x = 1$

$$\Rightarrow x = n\pi + \frac{\pi}{4}$$

$$\because \cos x < 0$$

$$\Rightarrow x = (2n+1)\pi + \frac{\pi}{4}; n \in \mathbb{I}$$

17. Consider the system of equations $\sin x \cos 2y = (a^2 - 1)^2 + 1$ and $\cos x \sin 2y = a + 1$. Which of the following ordered pairs (x, y) of real numbers can satisfy the given system of equations for permissible real values of a :

समीकरणों $\sin x \cos 2y = (a^2 - 1)^2 + 1$ तथा $\cos x \sin 2y = a + 1$ के निकाय पर ध्यान दें, वास्तविक संख्याओं का क्रमित युग्म (x, y) दिये गये समीकरणों के निकाय को a के सभी संभव मानों के लिए संतुष्ट करता है, होगा :

(A) $\left(\frac{-\pi}{2}, \frac{-\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ (C) $\left(\frac{3\pi}{2}, \frac{-\pi}{2}\right)$ (D) $\left(\frac{-\pi}{2}, \frac{3\pi}{2}\right)$

Ans. ACD

Sol. $\because \sin x \cos 2y = (a^2 - 1)^2 + 1$

$\because \sin x \cos 2y \in [-1, 1]$

$\& (a^2 - 1)^2 + 1 \in [1, \infty)$

$\Rightarrow (a^2 - 1)^2 + 1 = 1$

$\Rightarrow a^2 - 1 = 0$

$\Rightarrow a = \pm 1$

$\Rightarrow \sin x \cos 2y = 1$

$\Rightarrow \sin x = 1 \& \cos 2y = 1$

or $\sin x = -1 \& \cos 2y = -1$

$\Rightarrow x = (4n+1)\frac{\pi}{2} \& y = n\pi$

or $x = 2n\pi - \frac{\pi}{2} \& y = (2n+1)\frac{\pi}{2}$

$\Rightarrow (x, y) = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \left(-\frac{\pi}{2}, -\frac{\pi}{2}\right), \left(\frac{3\pi}{2}, -\frac{\pi}{2}\right), \left(-\frac{\pi}{2}, \frac{3\pi}{2}\right)$

18. If $(\cos^2 x + \sec^2 x)(1 + \tan^2 2y)(3 + \sin 3z) = 4$, then :

(A) x may be a multiple of π

(B) x cannot be an even multiple of π

(C) z can be a multiple of π

(D) y can be a multiple of $\frac{\pi}{2}$

यदि $(\cos^2 x + \sec^2 x)(1 + \tan^2 2y)(3 + \sin 3z) = 4$, तो :

(A) x, π का गुणक हो सकता है

(B) x, π का सम गुणक नहीं हो सकता

(C) z, π का गुणक हो सकता है

(D) y, $\frac{\pi}{2}$ का गुणक हो सकता है

Ans. AD

Sol. $\underbrace{\left(\cos^2 x + \frac{1}{\cos^2 x}\right)}_{\in [2, \infty)} \underbrace{(1 + \tan^2 2y)}_{[1, +\infty)} \underbrace{(3 + \sin 3z)}_{\in [2, 4]} = 4$

Only possibility are

$\cos^2 x + \frac{1}{\cos^2 x} = 2 \Rightarrow x = n\pi; n \in I$

$\& \sec^2 2y = 1 \Rightarrow y = \frac{n\pi}{2}; n \in I$

$\& 3 + \sin 3z = 2$

$\Rightarrow \sin 3z = -1$

$\Rightarrow z = \frac{n\pi}{3} + (-1)^n \frac{\pi}{2}; n \in I$

19. A solution (x, y) of the system of equations $x - y = 1/3$ and $\cos^2(\pi x) - \sin^2(\pi y) = 1/2$ is given by :

समीकरण निकाय $x - y = 1/3$ तथा $\cos^2(\pi x) - \sin^2(\pi y) = 1/2$ का हल (x, y) दिया जा सकता है :

- (A) $(7/6, 5/6)$ (B) $(2/3, 1/3)$ (C) $(-5/6, -7/6)$ (D) $(13/6, 11/6)$

Ans. ACD

Sol. $x - y = \frac{1}{3} \quad \dots(i)$

$$\cos^2 \pi x - \sin^2 (\pi y) = \frac{1}{2}$$

$$\Rightarrow \cos \pi (x + y) \cdot \sin \pi (x - y) = \frac{1}{2}$$

$$\Rightarrow \cos \pi (x + y) \sin \frac{\pi}{3} = \frac{1}{2} \quad \left\{ \because x - y = \frac{1}{3} \right\}$$

$$\Rightarrow \cos \pi (x + y) = 1$$

$$\Rightarrow x + y = \frac{2n\pi}{\pi}; n \in I$$

$$\Rightarrow x + y = 2n \quad \dots(ii)$$

from (i) & (ii)

$$x = n + \frac{1}{6} \quad \& \quad y = n - \frac{1}{6}$$

$$\text{For } n = -1 \quad x = -\frac{5}{6} \quad y = -\frac{7}{6}$$

$$n = 0 \quad x = \frac{1}{6} \quad y = -\frac{1}{6}$$

$$n = 1 \quad x = \frac{7}{6} \quad y = \frac{5}{6}$$

$$n = 2 \quad x = \frac{13}{6} \quad y = \frac{11}{6}$$

20. $\tan |x| = |\tan x|$ if:

$\tan |x| = |\tan x|$ यदि :

$$(A) x \in \left[-\frac{\pi(2k+1)}{2}, -\pi k \right]$$

$$(B) x \in \left[\pi k, \frac{\pi(2k+1)}{2} \right]$$

$$(C) x \in \left[-\pi k, -\frac{\pi(2k-1)}{2} \right]$$

$$(D) x \in \left[\frac{\pi(2k-1)}{2}, \pi k \right], k \in N$$

Ans. AB

Sol. $\tan |x| = |\tan x|$

Case I :- If $x \in [0, \infty)$

$$\Rightarrow \tan x = |\tan x|$$

If $\tan x \geq 0$, then $\underbrace{\tan x = \tan x}_{\text{always true}}$

If $\tan x < 0$ then $\underbrace{\tan x = 0}_{\text{Not possible}}$

$$\Rightarrow x \in \left[k\pi, (2k+1)\frac{\pi}{2} \right)$$

Case II :- If $x \in (-\infty, 0)$

$$-\tan x = |\tan x|$$

If $\tan x > 0$ then $\underbrace{\tan x = 0}_{\text{Not possible}}$

If $\tan x \leq 0$ then $-\tan x = -\tan x$

$$\Rightarrow x \in \left(-(2k+1)\frac{\pi}{2}, -\pi k \right)$$